

Supplementary Material

Insulin Resistance and Systemic Inflammation Below the HbA1c Prediabetes
Threshold:
Age-Dependent Associations in U.S. Adults, NHANES 2015–March 2020

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Table S1: Sensitivity and robustness analyses of age modification in the HOMA-IR–hs-CRP association, NHANES 2015–March 2020

Analysis	<i>n</i>	Interaction / test	Age-specific HOMA-IR coefficients	Interpretation
Primary model	3,079	$F(2, 40) = 5.69$, $p = 0.00671$	20–39: 0.0266 (0.0038, 0.0495) 40–54: 0.0511 (0.0132, 0.0890) 55+: −0.0189 (−0.0513, 0.0135)	Reference specification.
Raw hs-CRP outcome	3,079	$F(2, 40) = 3.96$, $p = 0.0269$	20–39: 0.127 (0.018, 0.235) 40–54: 0.160 (−0.056, 0.377) 55+: −0.109 (−0.282, 0.064)	Age interaction retained on the original outcome scale.
Log-transformed HOMA-IR	3,079	$F(2, 40) = 1.86$, $p = 0.169$	20–39: 0.127 (−0.0068, 0.261) 40–54: 0.172 (−0.0059, 0.350) 55+: −0.0046 (−0.153, 0.144)	Directional pattern remained, but formal age-interaction evidence was attenuated.
Raw hs-CRP + log HOMA-IR	3,079	$F(2, 40) = 1.43$, $p = 0.251$	20–39: 0.475 (−0.146, 1.096) 40–54: 0.338 (−0.621, 1.296) 55+: −0.165 (−0.937, 0.607)	Formal interaction not detected.
No continuous HbA1c adjustment	3,079	$F(2, 40) = 5.57$, $p = 0.00732$	20–39: 0.0284 40–54: 0.0521 55+: −0.0171	Essentially unchanged from the primary model.
BMI substituted for waist circumference ^a	3,072	$F(2, 40) = 4.91$, $p = 0.0124$	20–39: 0.0304 (0.0058, 0.0550) 40–54: 0.0601 (0.0234, 0.0968) 55+: −0.0085 (−0.0468, 0.0298)	Age pattern retained under alternative adiposity adjustment.
hs-CRP ≤ 10 mg/L, log outcome	2,895	$F(2, 40) = 2.58$, $p = 0.0883$	20–39: 0.0249 (0.0054, 0.0443) 40–54: 0.0478 (0.0118, 0.0838) 55+: −0.0042 (−0.0367, 0.0283)	Age-specific pattern remained, but formal interaction evidence weakened.
hs-CRP ≤ 10 mg/L, raw outcome	2,895	$F(2, 40) = 4.03$, $p = 0.0255$	20–39: 0.0824 (0.0300, 0.1348) 40–54: 0.0905 (0.0108, 0.1702) 55+: −0.0304 (−0.1077, 0.0469)	Interaction retained after restricting inflammatory extremes.

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Supplementary Table S1 continued

Analysis	<i>n</i>	Interaction / test	Age-specific HOMA-IR coefficients	Interpretation
HbA1c < 6.0%	4,018	$F(2, 40) = 5.74$, $p = 0.00644$	20–39: 0.0185 (−0.0034, 0.0404) 40–54: 0.0448 (0.0070, 0.0826) 55+: −0.0190 (−0.0457, 0.0078)	Relaxing eligibility did not remove age modification.
HbA1c < 6.5%	4,492	$F(2, 40) = 5.58$, $p = 0.00731$	20–39: 0.0120 (−0.0101, 0.0341) 40–54: 0.0371 (0.0046, 0.0696) 55+: −0.0290 (−0.0568, −0.0011)	Age modification persisted with a broader glycemic range.
No HbA1c cutoff ^b	4,639	$F(2, 40) = 5.74$, $p = 0.00643$	20–39: 0.0115 (−0.0081, 0.0311) 40–54: 0.0349 (0.0065, 0.0633) 55+: −0.0212 (−0.0456, 0.0032)	Simple HbA1c-threshold selection did not explain older attenuation.
Continuous linear age interaction	3,079	$F(1, 40) = 3.93$, $p = 0.0544$	Not represented by three age-group coefficients.	Borderline evidence under a strictly linear age-varying slope.
Continuous quadratic age interaction	3,079	$F(2, 40) = 3.43$, $p = 0.0421$	Not represented by three age-group coefficients.	Evidence of nonlinear age modification; quadratic component alone $p = 0.0857$.
Natural cubic spline age interaction	3,079	$F(4, 40) = 3.09$, $p = 0.0262$	Not represented by three age-group coefficients.	Flexible continuous-age model supported age-dependent variation.
Leave-one-survey-cluster-out ^c	3,023– 3,063	79 refits	Not applicable.	Interaction F ranged from 3.59 to 7.66 and p from 0.00157 to 0.0369. No single masked stratum–PSU cluster eliminated the interaction.

Notes: Values in parentheses are 95% confidence intervals. For models with ln(hs-CRP) as the outcome, coefficients are on the log outcome scale; raw-outcome coefficients are in hs-CRP units. All analyses used the NHANES complex survey design with duration-adjusted fasting-subsample weights, masked variance strata, and masked variance units. Sensitivity analyses were performed to evaluate robustness and were not treated as additional primary hypothesis tests.

^a To isolate the adiposity specification itself, the BMI analysis was restricted to participants in the primary analytic sample who also had nonmissing BMI.

^b Participants with diagnosed diabetes remained excluded; only the HbA1c eligibility threshold was removed.

^c Each refit deleted one unique masked stratum–PSU combination from the locked primary model.

Table S2: Sensitivity of age-specific HOMA-IR–hs-CRP associations to alternative adiposity specifications.

Adiposity measure	Age 20–39 years	Age 40–54 years	Age ≥ 55 years	Interaction	
	β (95% CI)	β (95% CI)	β (95% CI)	<i>F</i>	<i>p</i>
Panel A. Full-period common sample ($n = 3,072$)					
Waist circumference	0.0265 (0.0038, 0.0493)	0.0517 (0.0137, 0.0897)	−0.0187 (−0.0512, 0.0139)	5.71	0.00661
BMI	0.0304 (0.0058, 0.0550)	0.0601 (0.0234, 0.0968)	−0.0085 (−0.0468, 0.0298)	4.91	0.0124
Waist-to-height ratio	0.0252 (0.0029, 0.0475)	0.0515 (0.0143, 0.0888)	−0.0244 (−0.0572, 0.0084)	7.13	0.00225
Panel B. WHR-available period common sample ($n = 1,899$)					
Waist circumference	0.0275 (0.0059, 0.0490)	0.0388 (−0.0062, 0.0838)	−0.0090 (−0.0530, 0.0351)	1.96	0.161
BMI	0.0324 (0.0099, 0.0549)	0.0485 (0.0040, 0.0929)	−0.0046 (−0.0537, 0.0445)	2.14	0.139
Waist-to-height ratio	0.0282 (0.0069, 0.0496)	0.0364 (−0.0053, 0.0781)	−0.0135 (−0.0553, 0.0282)	2.80	0.0799
Waist-to-hip ratio	0.0677 (0.0337, 0.1016)	0.0882 (0.0264, 0.1500)	0.0370 (−0.0239, 0.0979)	1.49	0.245

Notes: Values are survey-weighted age-specific coefficients (β) with 95% confidence intervals from models with log-transformed hs-CRP as the outcome. Each model replaces the primary adiposity covariate with the anthropometric measure indicated while retaining the remaining primary adjustment covariates. Panel A restricts all three specifications to the same participants with complete waist circumference, BMI, and waist-to-height ratio data across NHANES 2015–March 2020, thereby holding sample composition constant across models. Panel B restricts all four specifications to the same participants during the survey period in which hip circumference was available, permitting direct comparison with waist-to-hip ratio while holding sample composition constant. Interaction tests evaluate the joint HOMA-IR-by-age-group terms. Denominator degrees of freedom were 40 in Panel A and 25 in Panel B. HOMA-IR, Homeostatic Model Assessment of Insulin Resistance; hs-CRP, high-sensitivity C-reactive protein; BMI, body mass index; WHR, waist-to-hip ratio.

Selection of the NHANES 2015-March 2020 analytic population

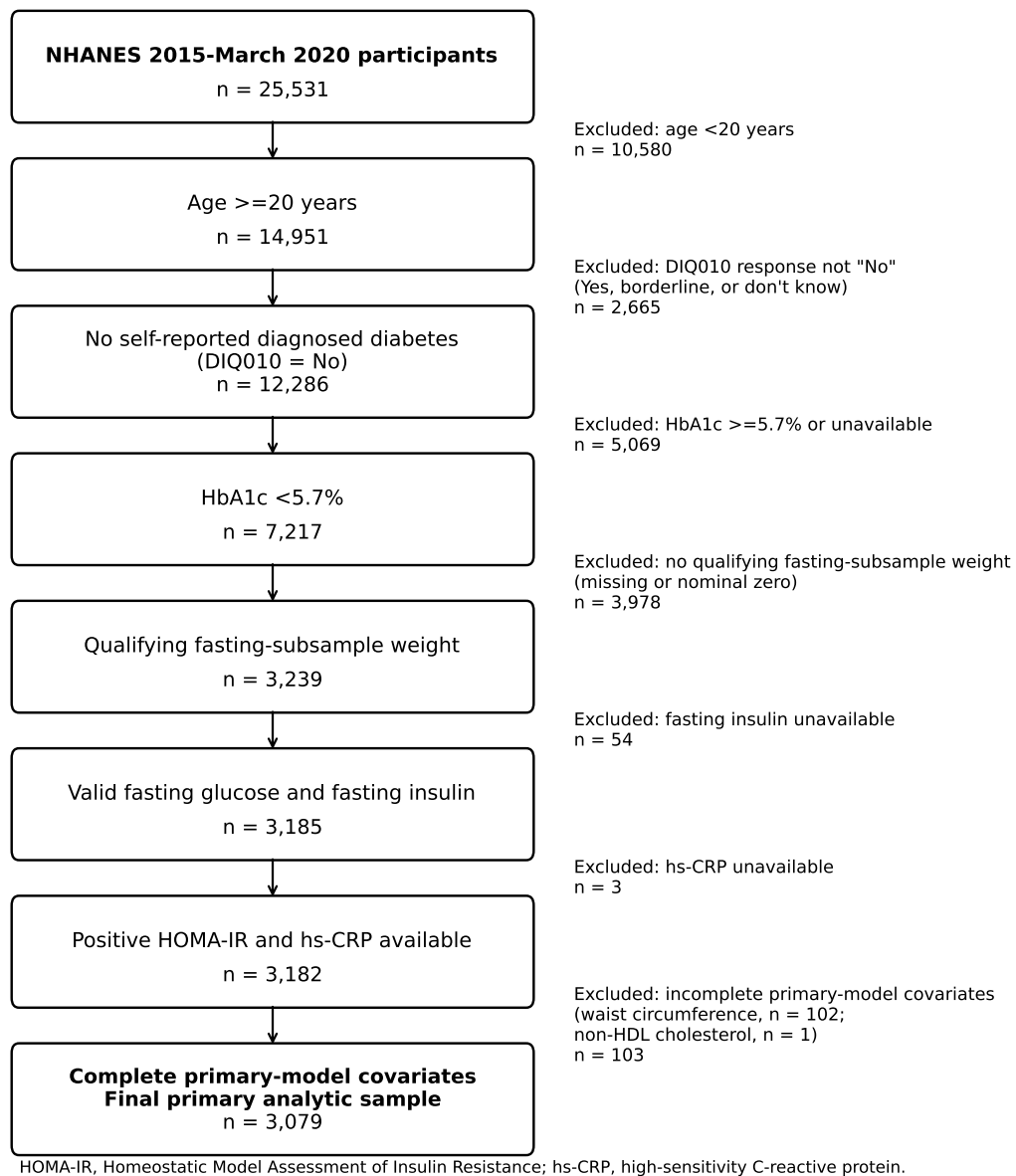


Figure S1: Selection of the NHANES 2015–March 2020 analytic population.